Practice and Experience in JHPC-Quantum: HPC Oriented quantum-HPC integration

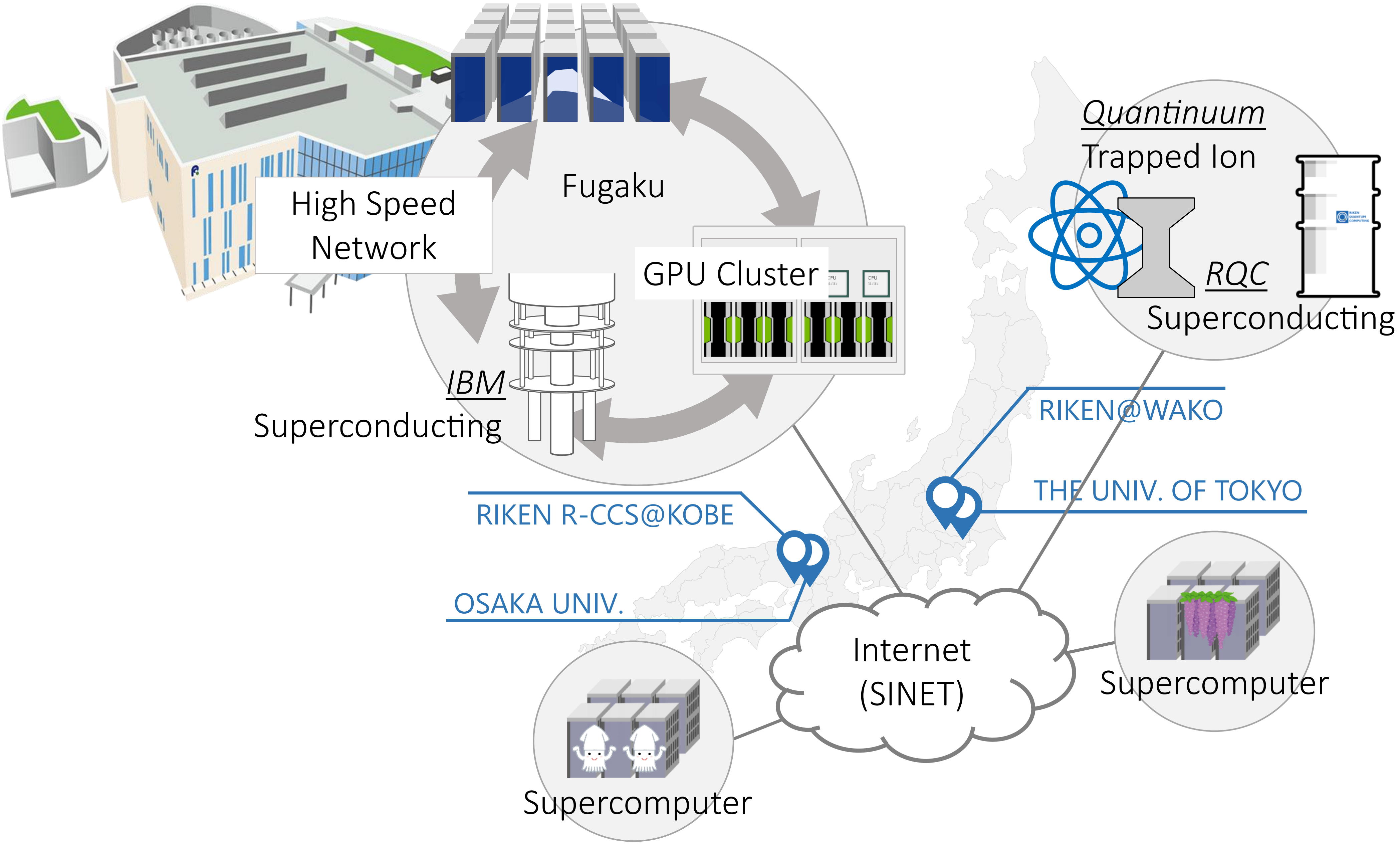
Miwako TSUJI

RIKEN Center for Computational Science (R-CCS)

Center for Computational Sciences (CCS) , University of Tsukuba

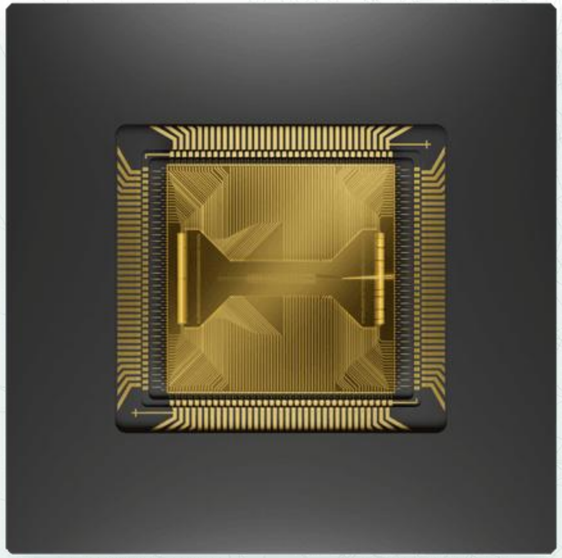
12th Jan. 2026

Overview (JHPC-Quantum & TRIP)



Recent Status of Quantum Systems

	IBM Kobe	Quantinuum 黎明 Reimei	RIKEN 叡 A
Name	Heron	System Model H1	A
Type	Superconductive	Trapped Ion	Superconductive
Qubit	156	20	64
Status	Running	Running	Running




Pictures
https://jp.newsroom.ibm.com/2023-12-05-IBM-Debuts-Next-Generation-Quantum-Processor-IBM-Quantum-System-Two,-Extends-Roadmap-to-Advance-Era-of-Quantum-Utility
https://www.quantinuum.com/products-solutions/quantinuum-systems/system-model-h1
https://www.riken.jp/medialibrary/riken/pr/news/2023/20231005_1_photo1.jpg

“A” RIKEN Superconductive Quantum Computer

- RIKEN Center for Quantum Computing
  - The successor of the *first* developed superconducting qubit
  - The cloud service started in Mar. 2023
  - 16 → 64 (now) → 144 qubits
    → 256 qubits (Fujitsu)

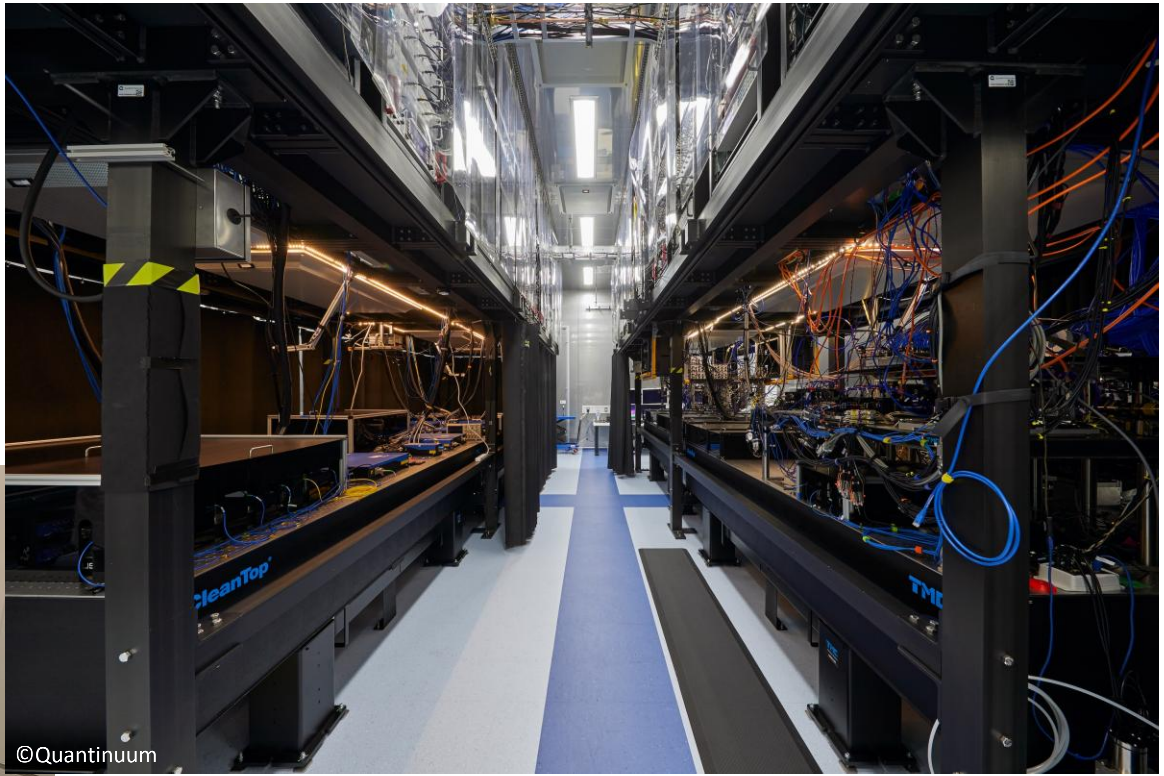
https://www.riken.jp/pr/news/2023/20231005_1/index.html

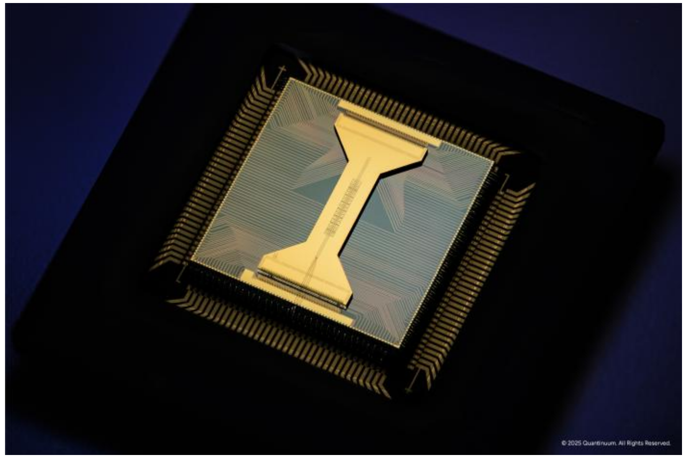
“Reimai” Quantinuum Trapped-Ion Quantum Computer

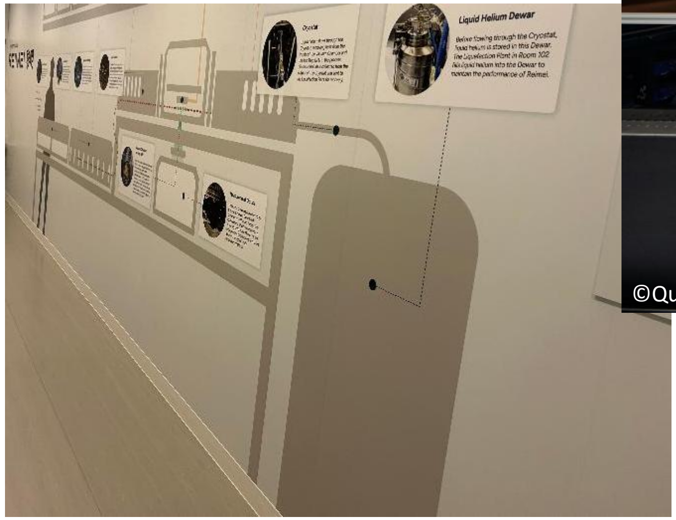
黎明

QUANTINUUM
REIMEI

Quantinuum Trapped-Ion Quantum Computer H1 (20 qubits) was installed in Wako Campus, Japan and is operated since Feb, 2025







IBM Superconductive Quantum Computer

- IBM Heron
  - 156-qubit

Quantum HPC related Projects in RIKEN R-CCS

TRIP3 (2024.04-

Expansion of Computationally Viable Region, Transformative Research Innovation Platforms of RIKEN Platforms,

A cross-disciplinary project to establish connections among RIKEN's cutting-edge research platforms such as *supercomputers*, *quantum computing*, large synchrotron radiation facilities, bioresource-projects.
R-CCS and Center for Quantum Computing (RQC) are tasked with "the value creation by quantum-HPC hybrid computing for Accelerating Research DX".

JHPC-Quantum (2024.11-

Research and Development of Quantum-Supercomputers Hybrid Platform for Exploration of Uncharted Computable Capabilities

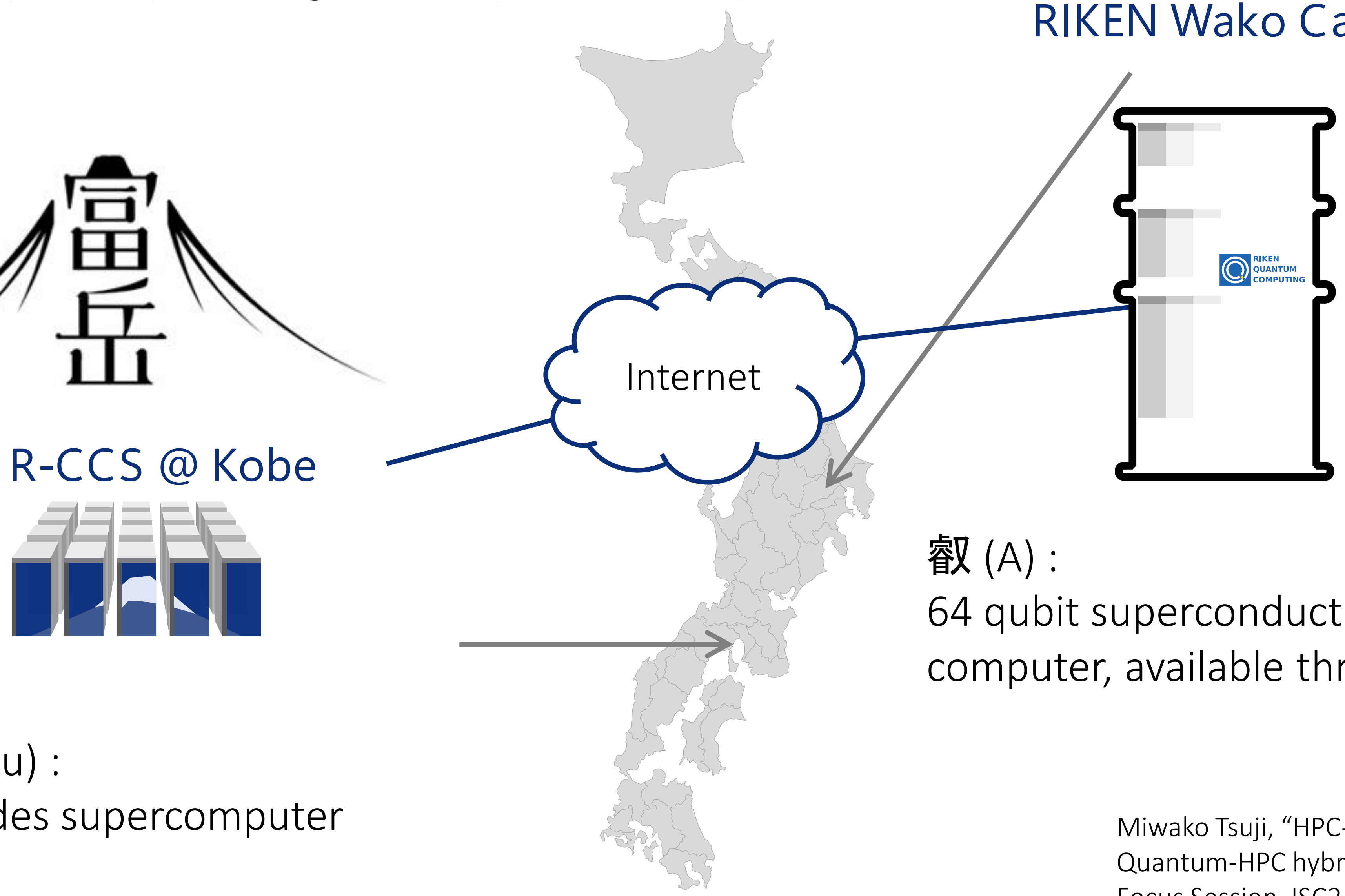
One of research projects in "Research and Development Project of the Enhanced Infrastructures for Post-5G Information and Communication Systems" in NEDO, an agency under the Ministry of Economy, Trade and Industry.
R-CCS, Softbank Corp., the university of Tokyo, and Osaka university
A platform that connects supercomputers and different types of quantum computers
to promote the early commercialization of quantum computers.

TRIP Proof of concept in 2023/2024 experiments and lessons learned

- Connect supercomputer **Fugaku** and quantum computer **A**

RIKEN Wako Campus @ Kobe

RIKEN R-CCS @ Kobe



富岳 (Fuagku) :
158,976 nodes supercomputer

叡 (A) :
64 qubit superconducting quantum computer, available through the cloud

Miwako Tsuji, “HPC-oriented design of Quantum-HPC hybrid computing platform“ Focus Session, ISC24, 2024

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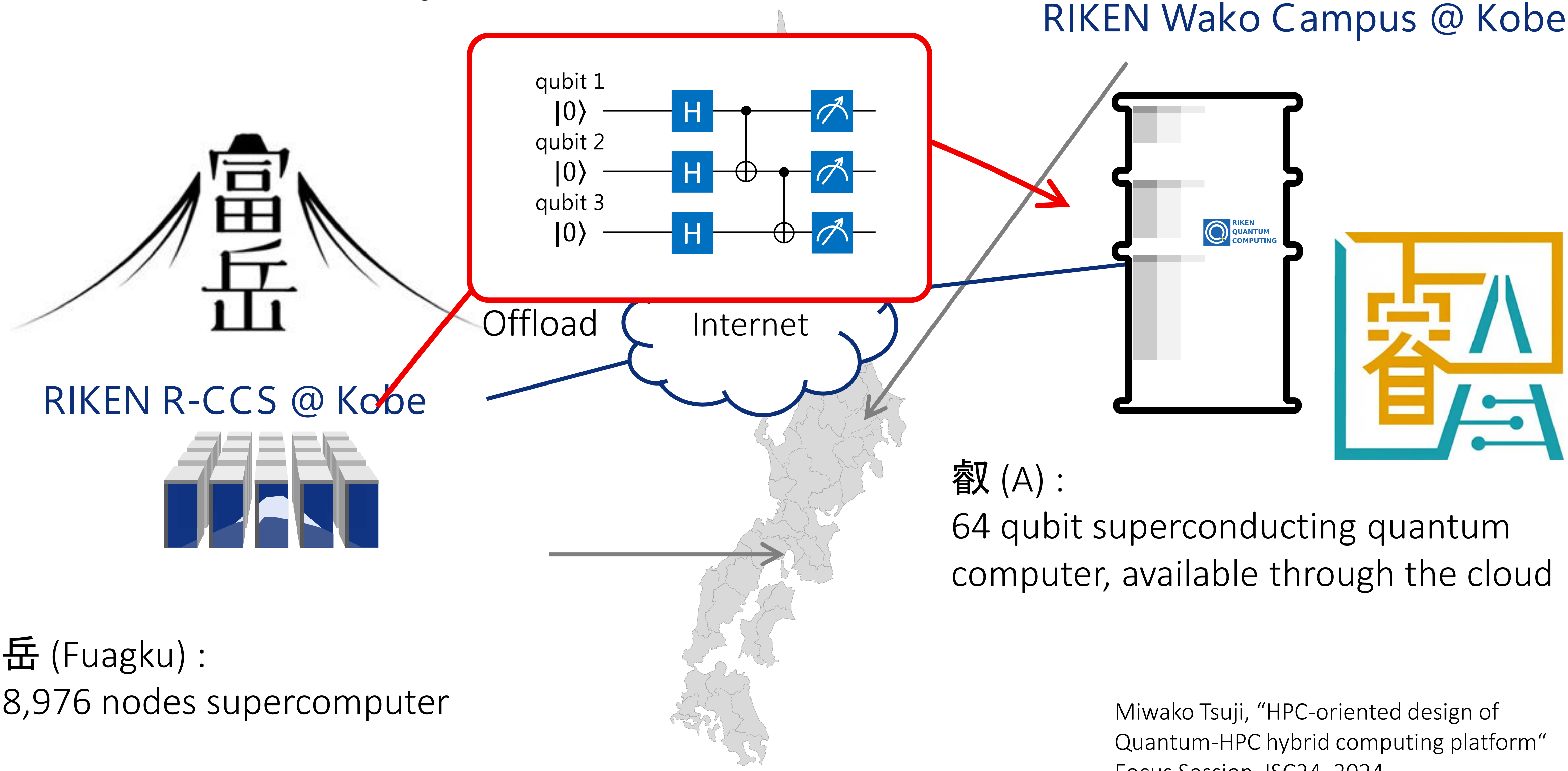

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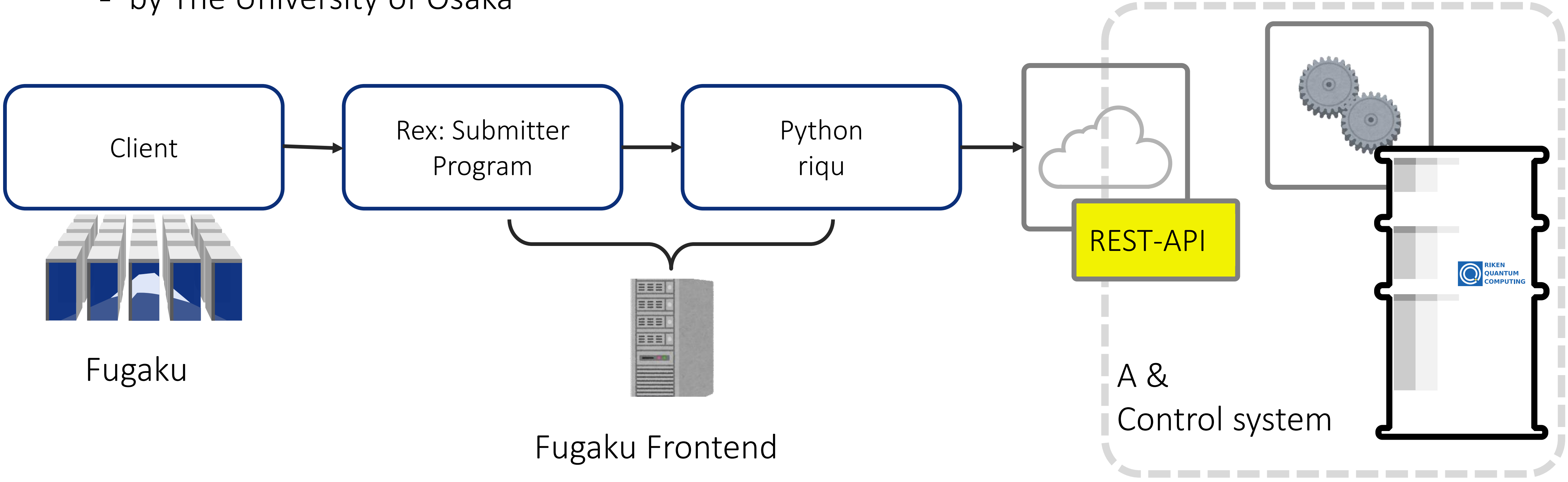
1st Experiment: RPC though a Python based library

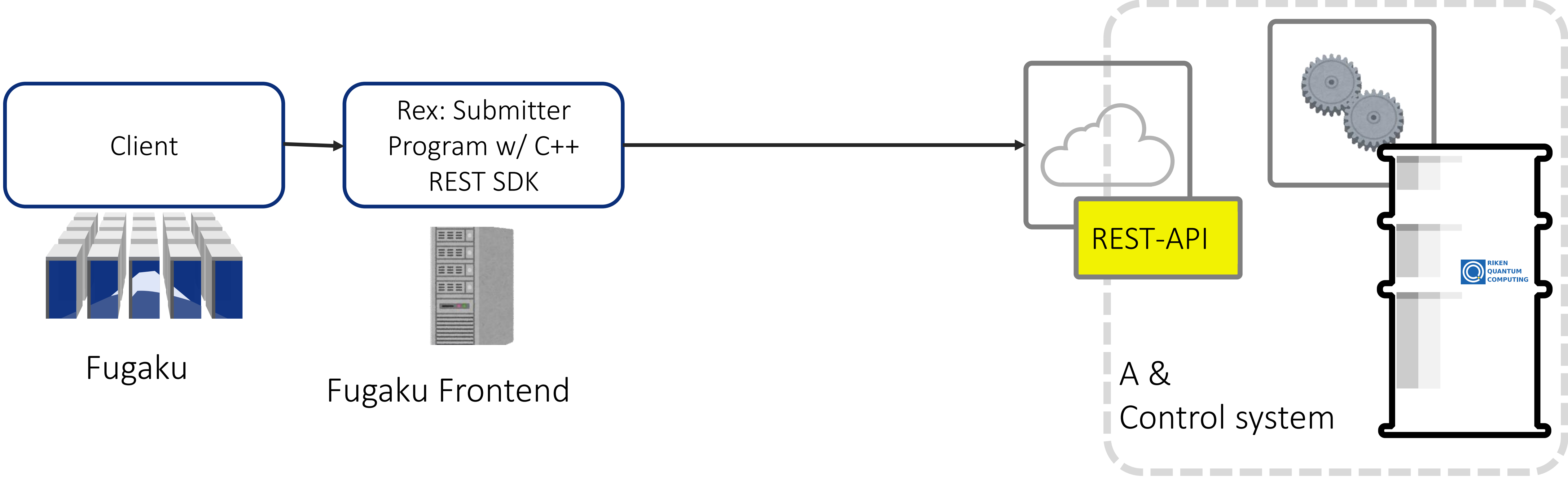
- A client program calls a remote submitter program
- The submitter program calls a Python library designed to use “A” through REST-API
- QURI Parts riqu: **R**EST **I**nterface for **Q**uantum **C**omputing
    - by The University of Osaka



Note: 4 of 64 qubit are available for us

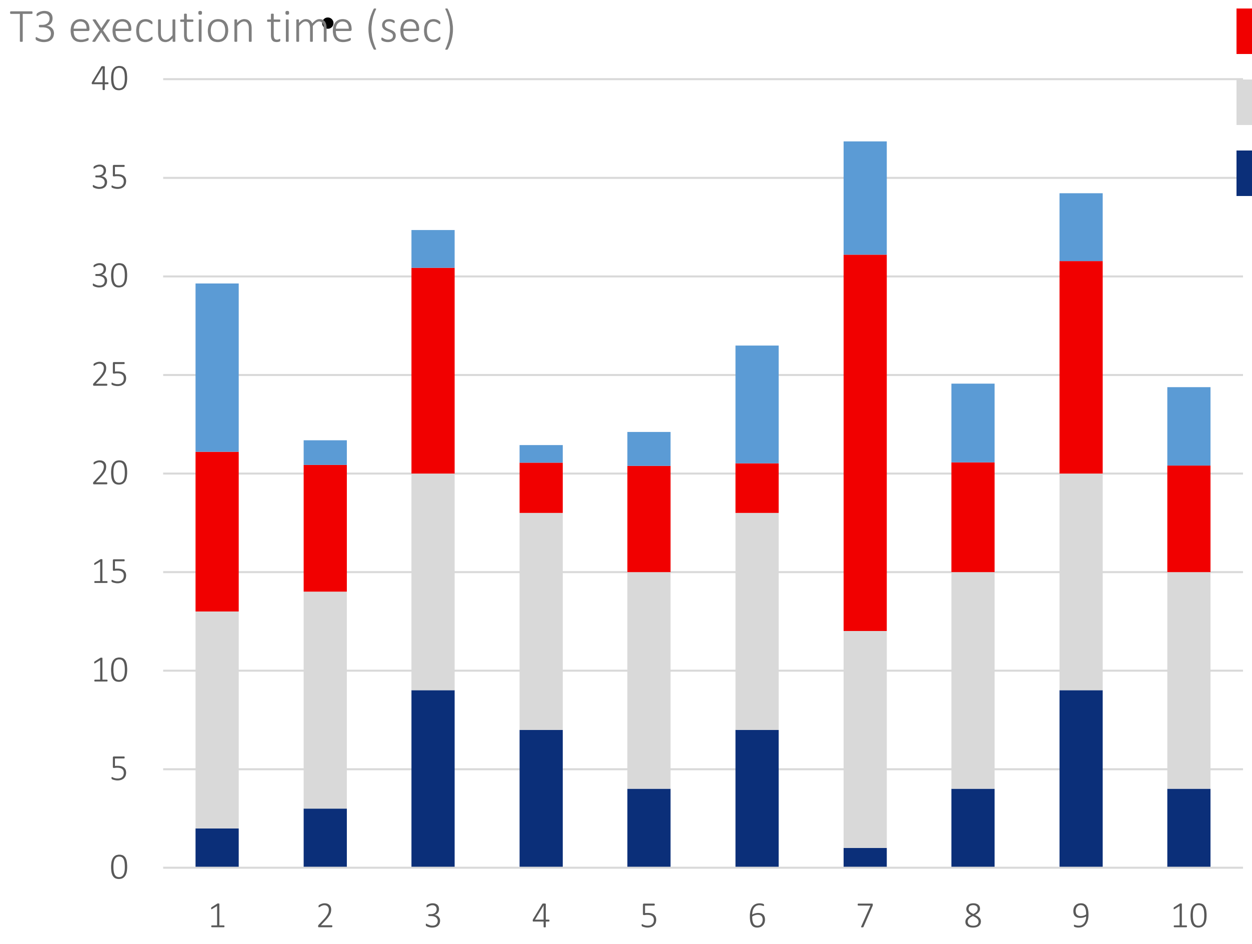
2nd Experiment: RPC though C++ based library

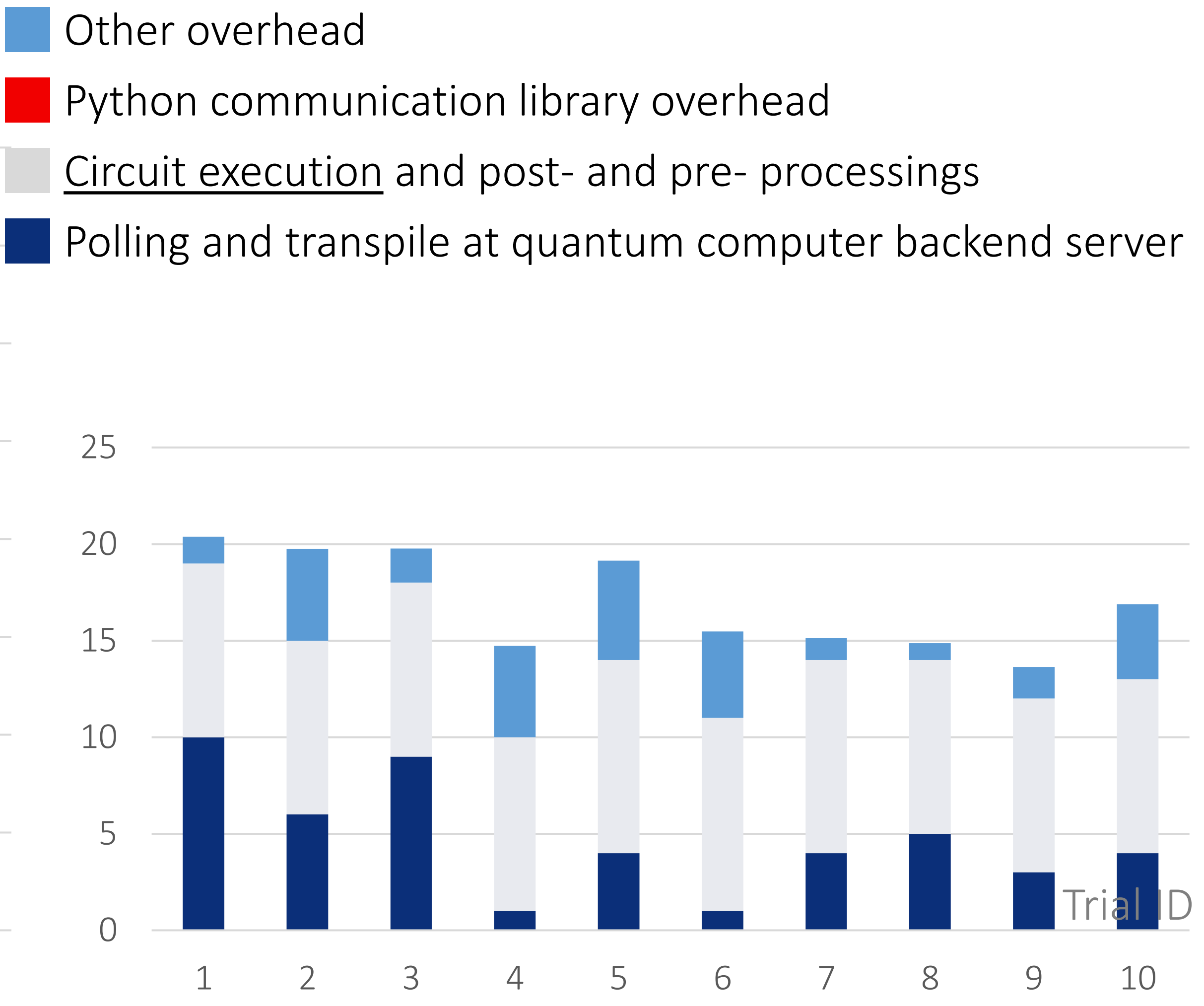
- The C++ REST SDK
  - https://github.com/microsoft/cpprestsdk



Result

- Quantum Circuit Encoding Workflow
- A simple step-by-step workflow
- Automatic quantum circuit encoding (AQCE) of a given arbitrary quantum state
  - Generate a circuit to make a gray scale image:
  - 4qubits, 4x4=16pixel





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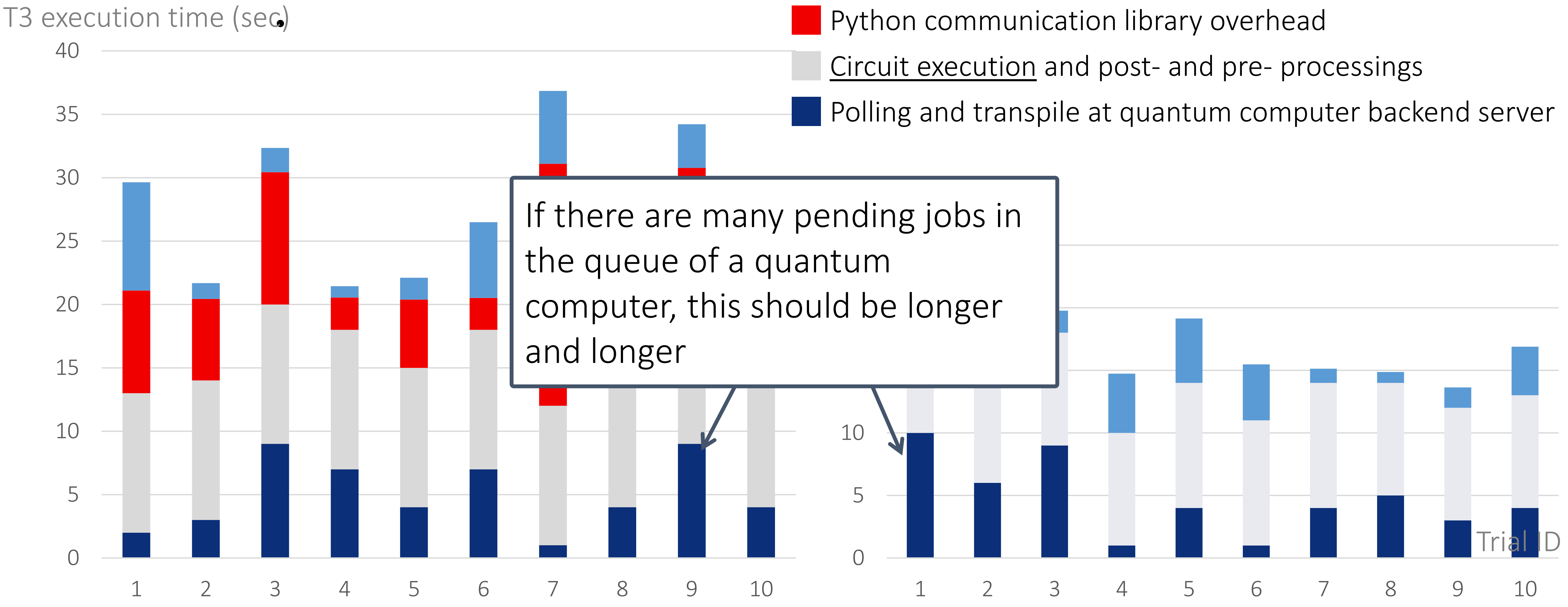

TRIP Proof of concept in 2023/2024 experiments and lessons learned

- We found our programming model cannot be used in our supercomputer system (esp “Fugaku”)

- Coordinated scheduling with quantum computer and supercomputer is required.
- Execution of HPC programs running multiple nodes should not be idle as possible as it can.

- Major programming environments of quantum computer program development are Python-based frameworks.
    - Python is difficult to be combined with HPC programs such as MPI, multithreaded programs in C/C++ and Fortran.
    - Recently, some quantum computing programming frameworks in C/C++ are proposed

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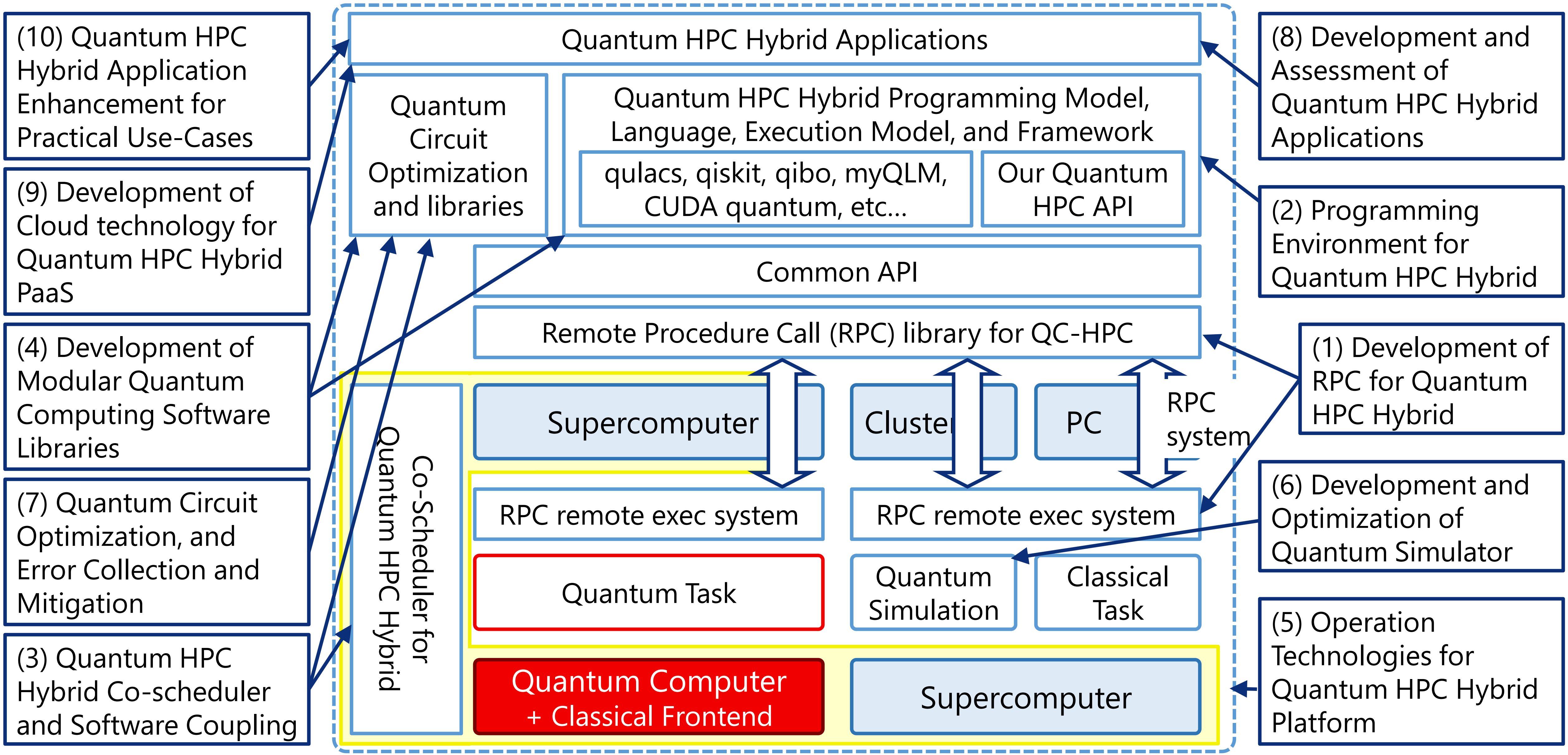
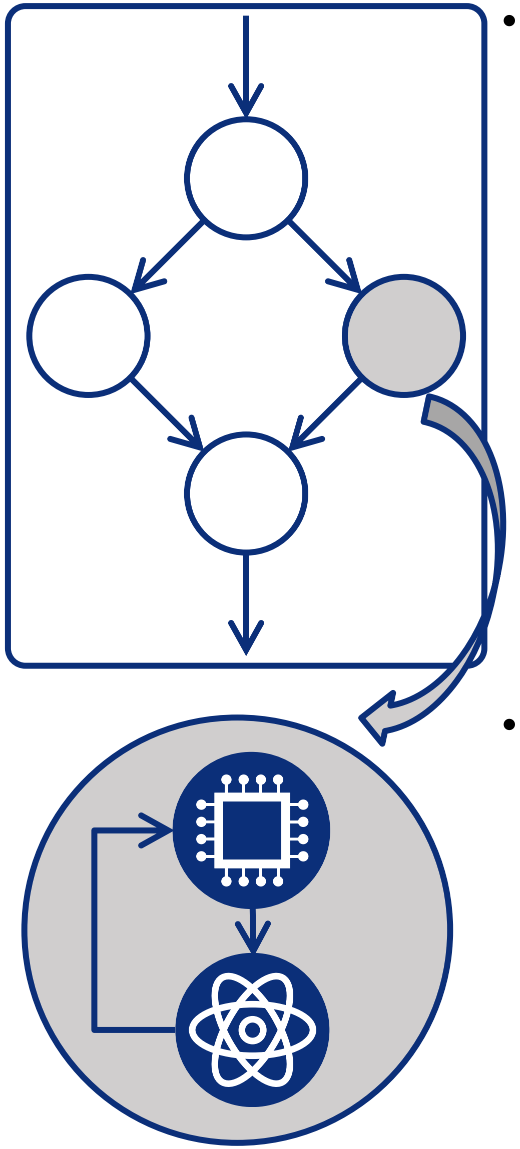
Research Items in JHPC-Quantum Project



Quantum HPC Middleware: Two Level Programming Model

- Workflow & Task based parallelism



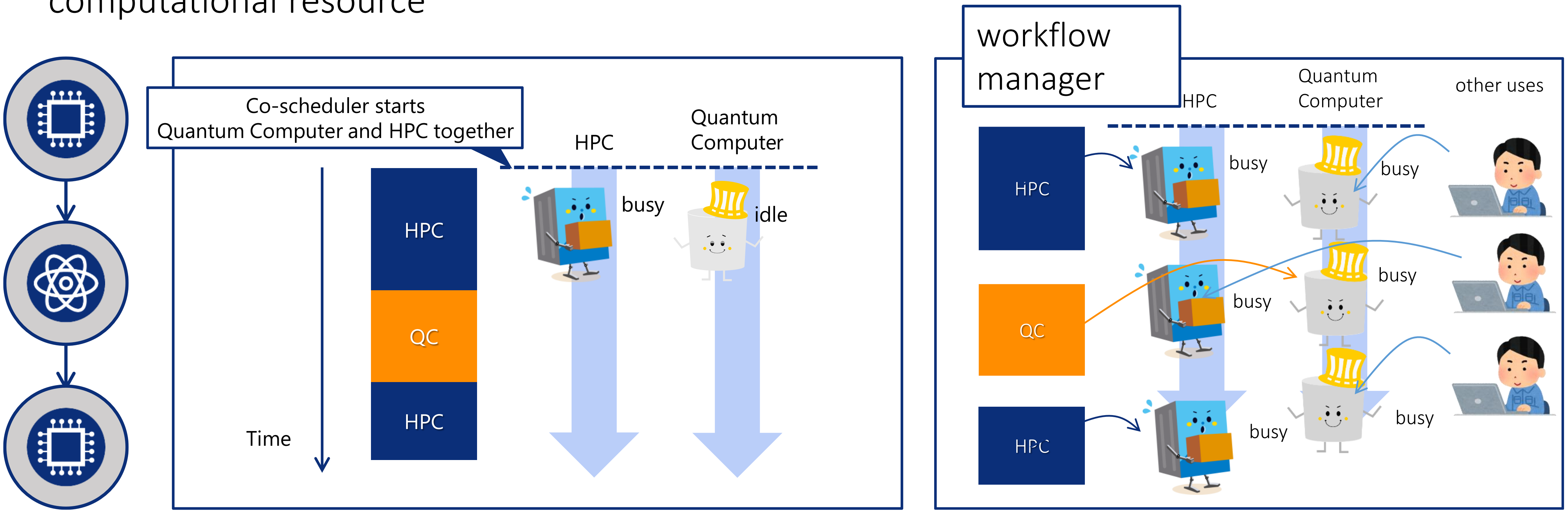
- <u>Workflow</u>
  - Tasks (different kernels) are managed by a workflow engine
  - The tasks are "**jobs**" in supercomputers and/or quantum computer
  - The workflow engine submit each of jobs in a certain order based on the descriptions about dependencies between jobs

Simple HPC Job	Simple QC Job	HPC-QC Job
running a kernel only on HPC	offloads a quantum circuit to quantum computer	includes both of HPC and quantum kernels

- <u>Task based Parallelism in HPC-QC Jobs</u>
  - offloads quantum kernel(s) to a quantum computer via remote procedure call (RPC)
  - A single HPC-QC job may offload multiple kernels
  - RPC is an asynchronous mechanism; Other classical kernels will be executed on CPU simultaneously

Why do we need the 1st Layer, Job-Scheduler Based Workflow?

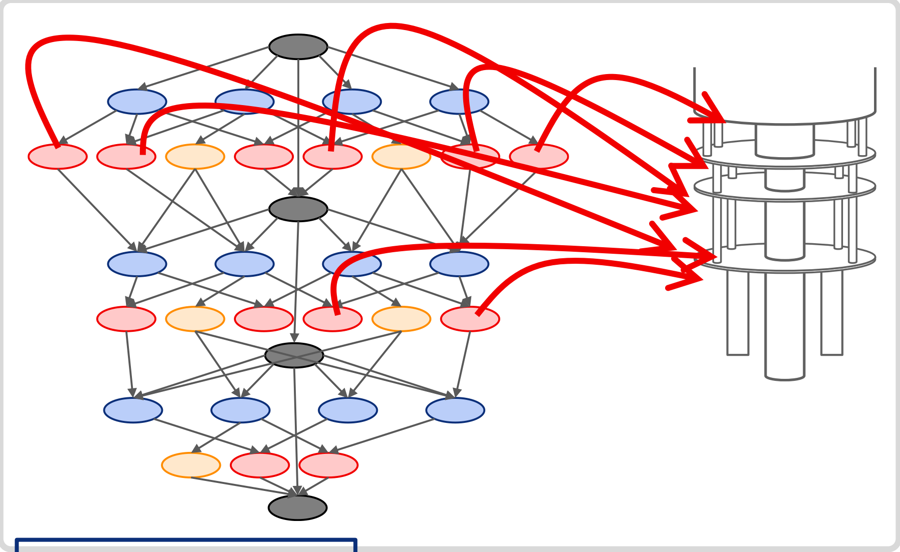
- Avoid the waiting times in quantum computer and supercomputer
  - Uses separate quantum kernels and HPC kernels in their applications
  - Define the order of their execution
- Job-schedulers in both computers may arrange jobs to maximize the throughput in their systems.
- A job may have to wait the result from another job, but the waiting time does not consume any computational resource

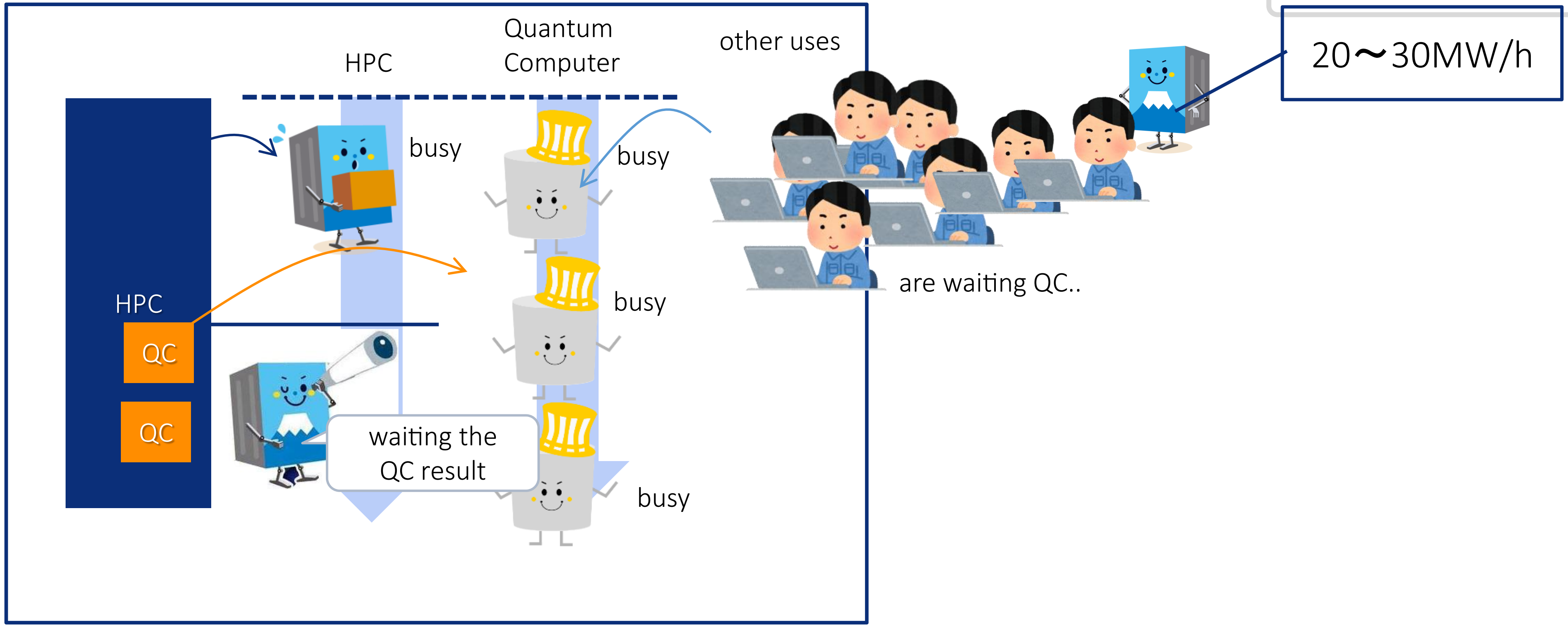


Why do we need 2nd Layer, task-based programming

- Tightly coupled quantum and HPC kernels in a single job
  - Difficult to separate quantum and HPC kernels completely
- Multiple quantum kernel calls from HPC in a short time (ex. VQE)
  - Difficult to wait each of the quantum circuit executions
- (Still) We must avoid the waiting time in HPC,

especially for large scale HPC jobs





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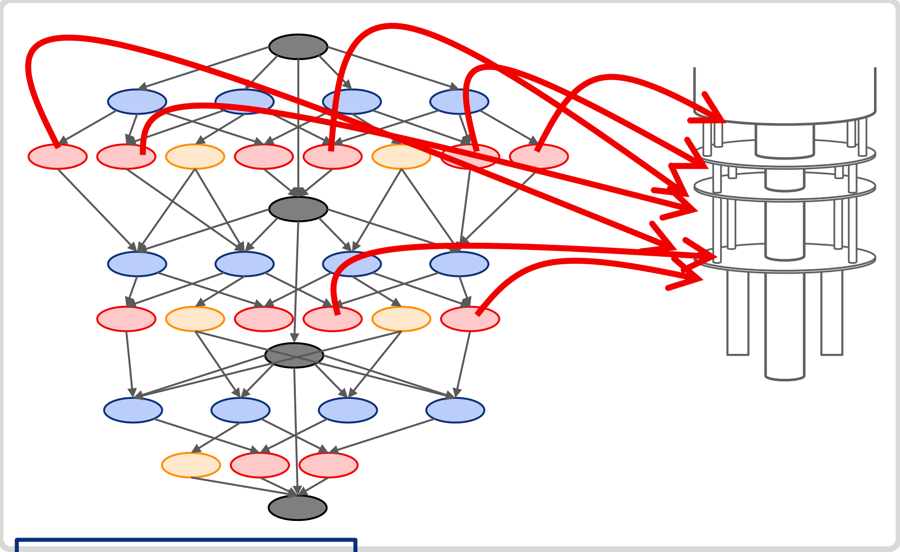
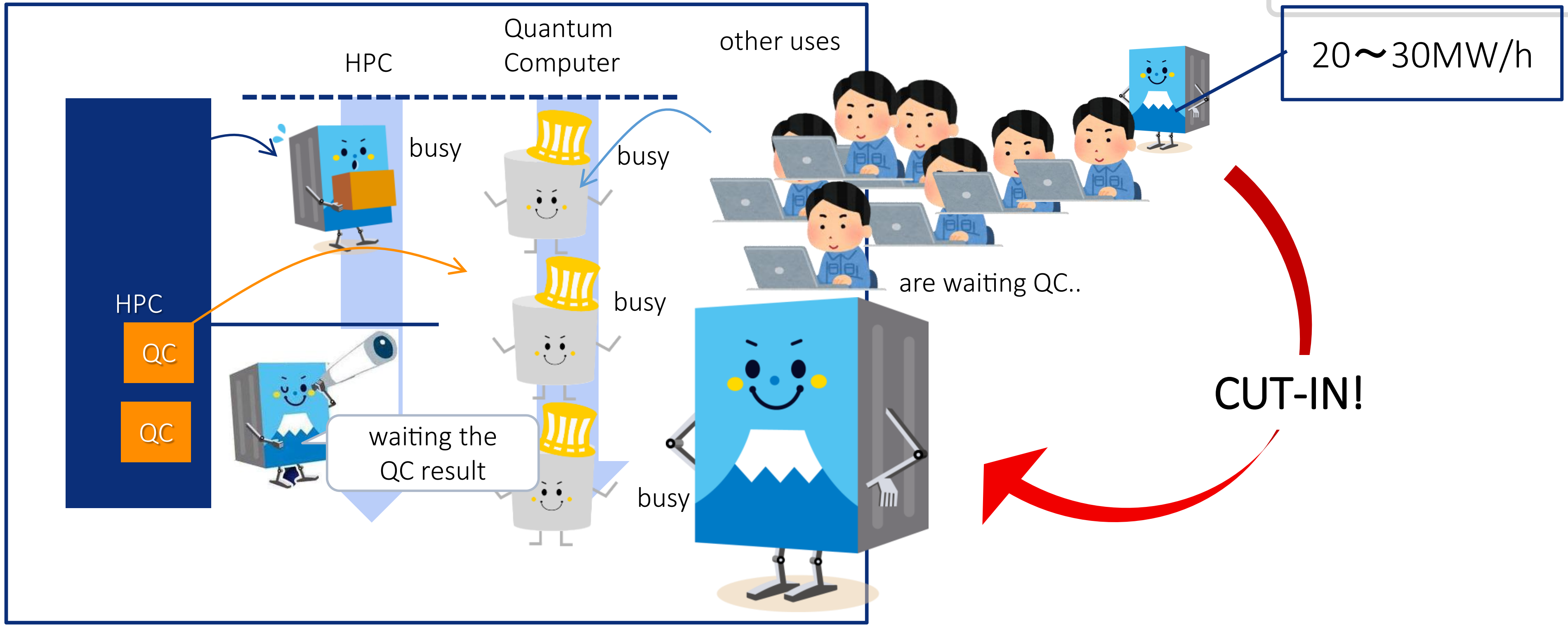
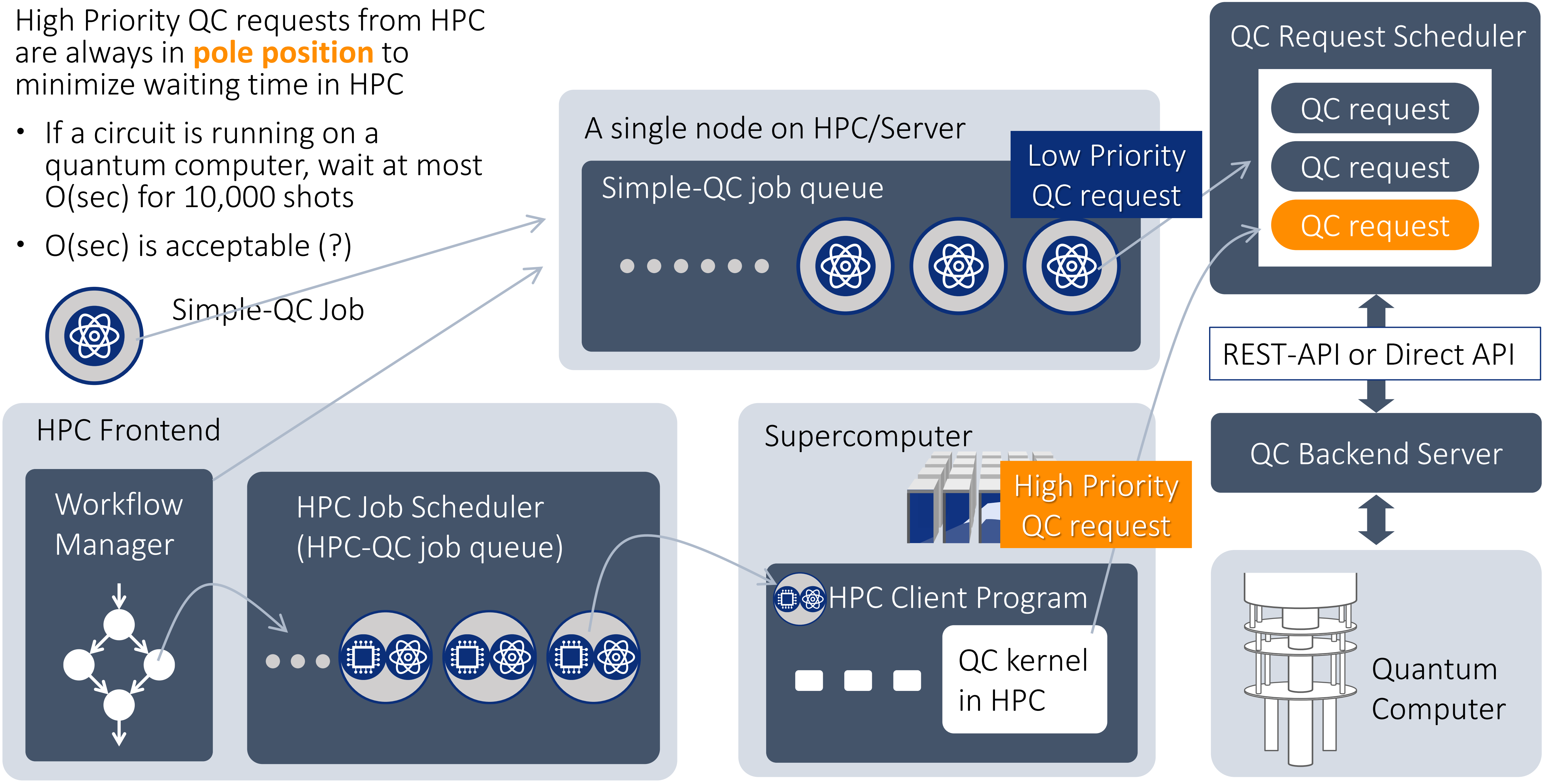




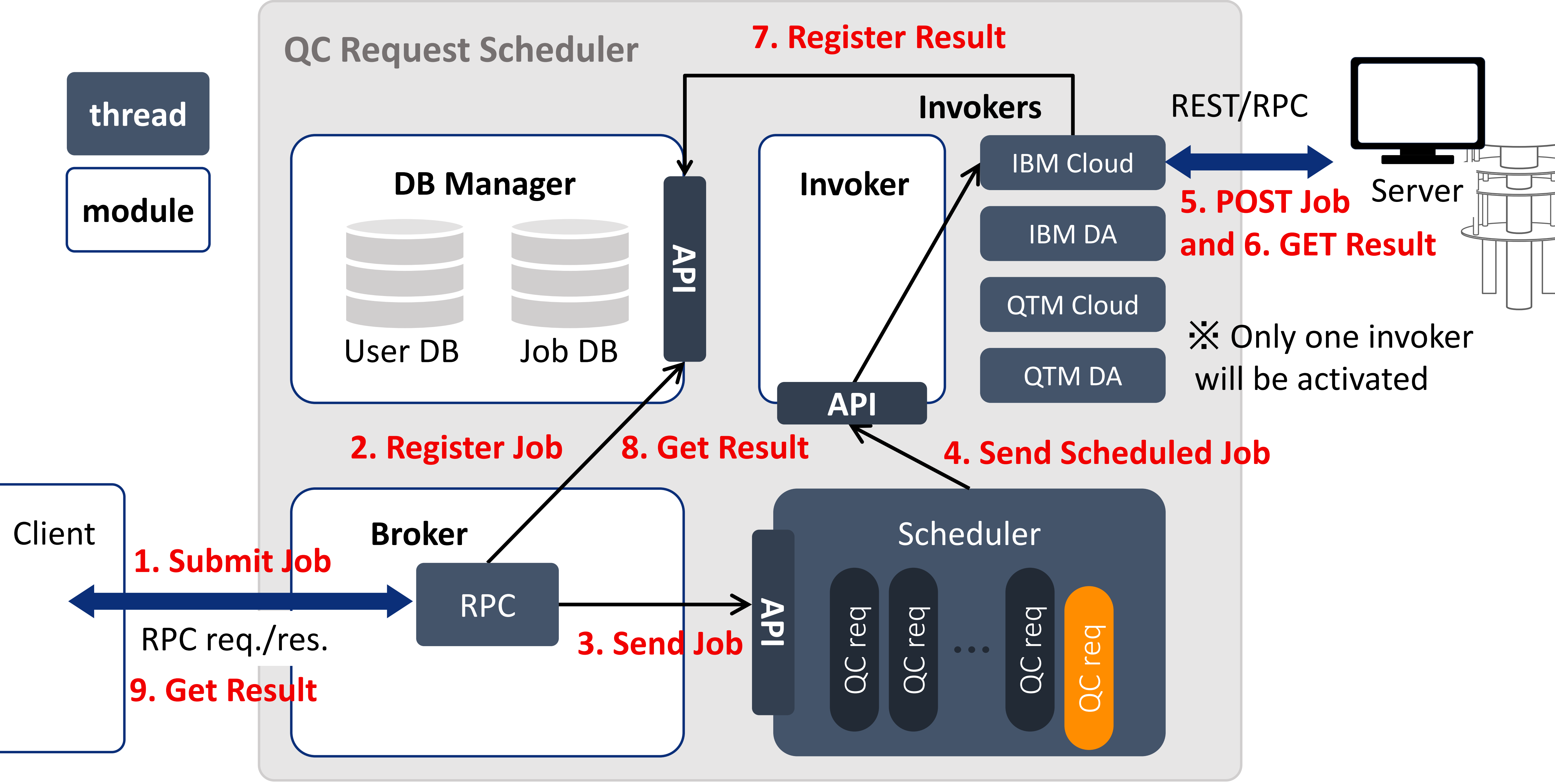
Execution Environment

High Priority QC requests from HPC are always in **pole position** to minimize waiting time in HPC

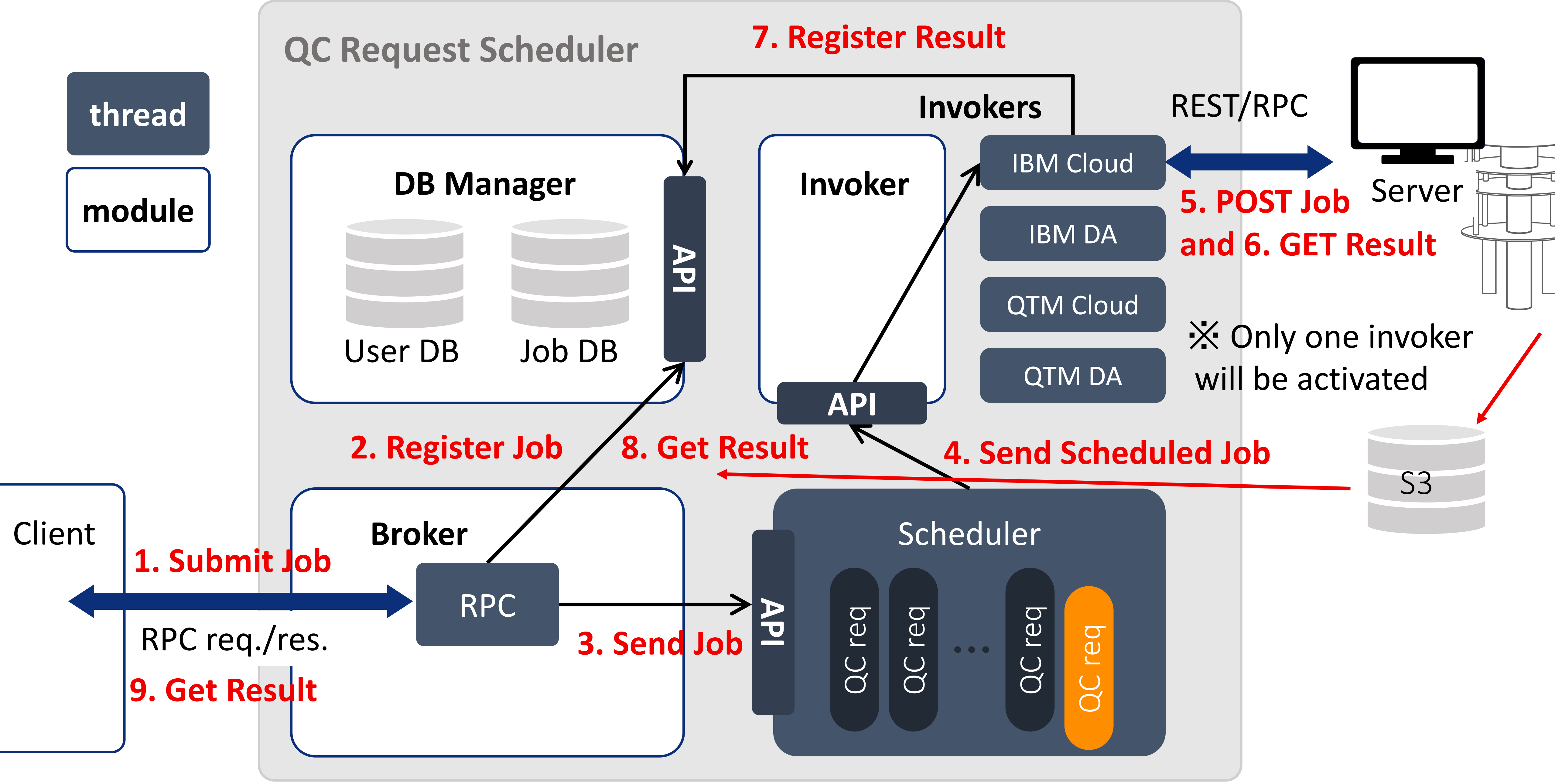
- If a circuit is running on a quantum computer, wait at most O(sec) for 10,000 shots
- O(sec) is acceptable (?)



SQC (Supercomputer Quantum computer Computation) Scheduler

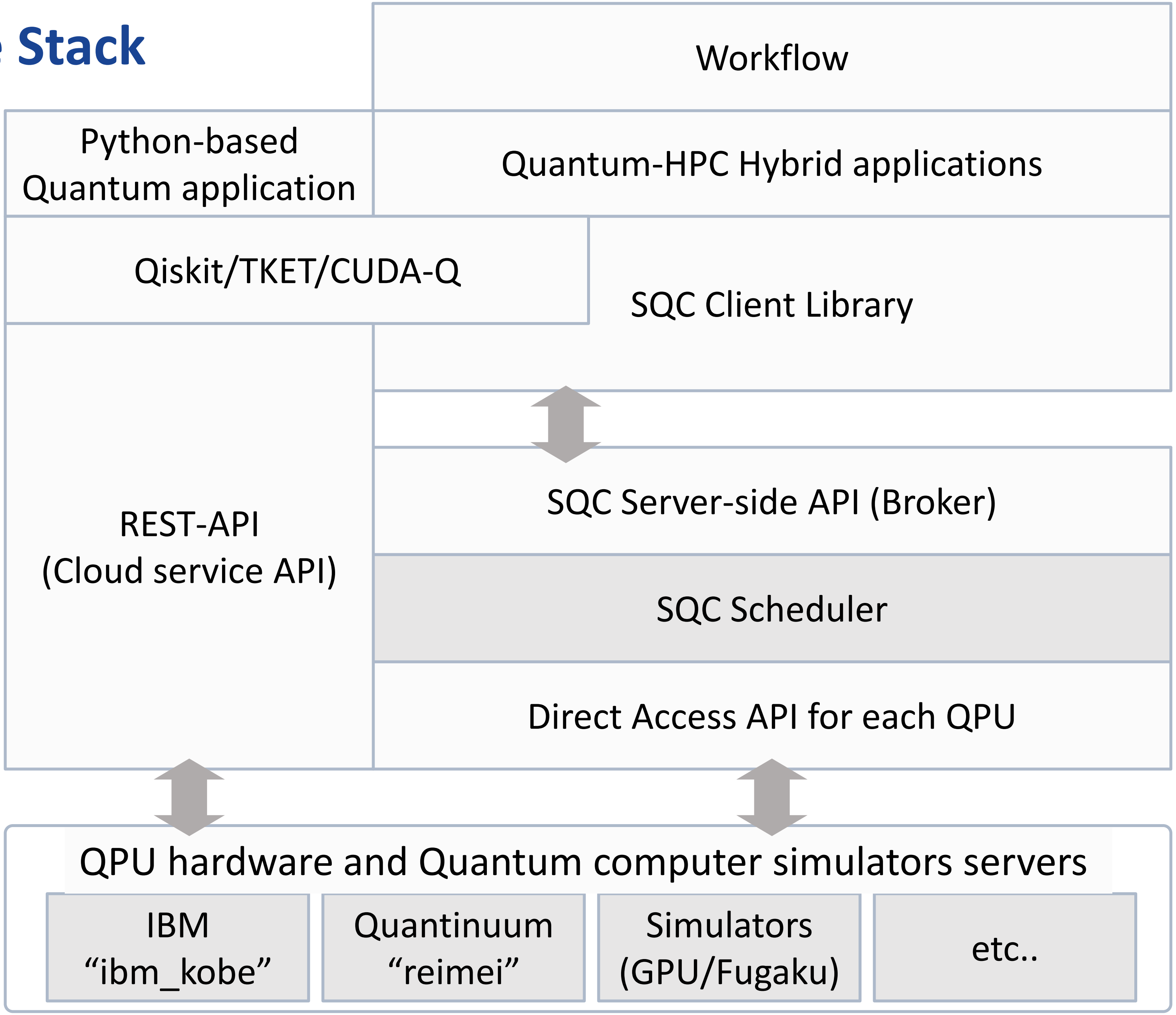


SQC Scheduler for IBM_Kobe



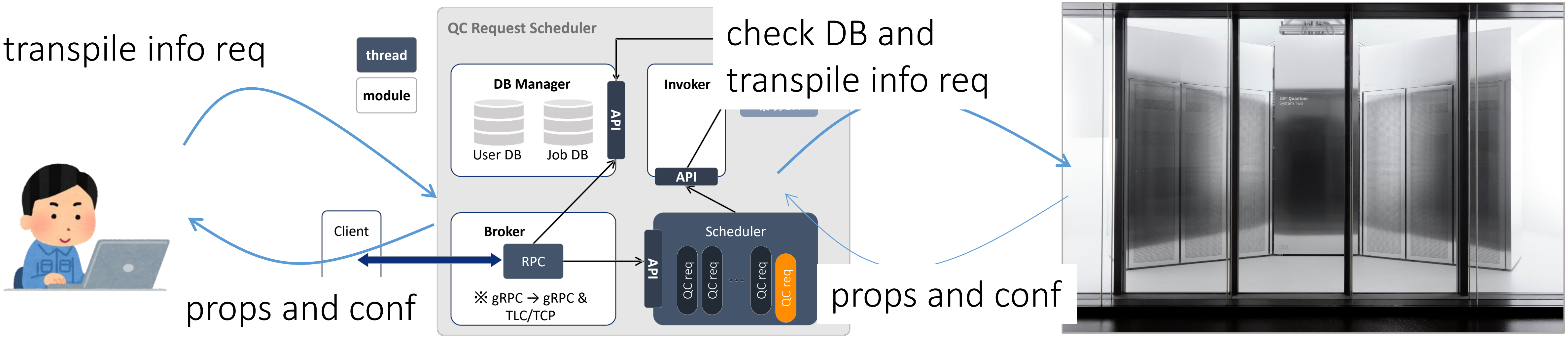
JHPC-Quantum Software Stack

- Two ways to access QC
  - Our SQC APIs and RPC. The RPC uses direct access APIs provided by QPU venders
  - REST-APIs through vender clouds
- Workflow tools
  - Tierkreis (Quantinuum), Prefect, Xcrypt
- QC software libraries
  - JHPC Quantum QC software lib
  - Qunasys QURI SDK
  - Error mitigation
  - Error suppression



Additional API for ibm_kobe, Transpile

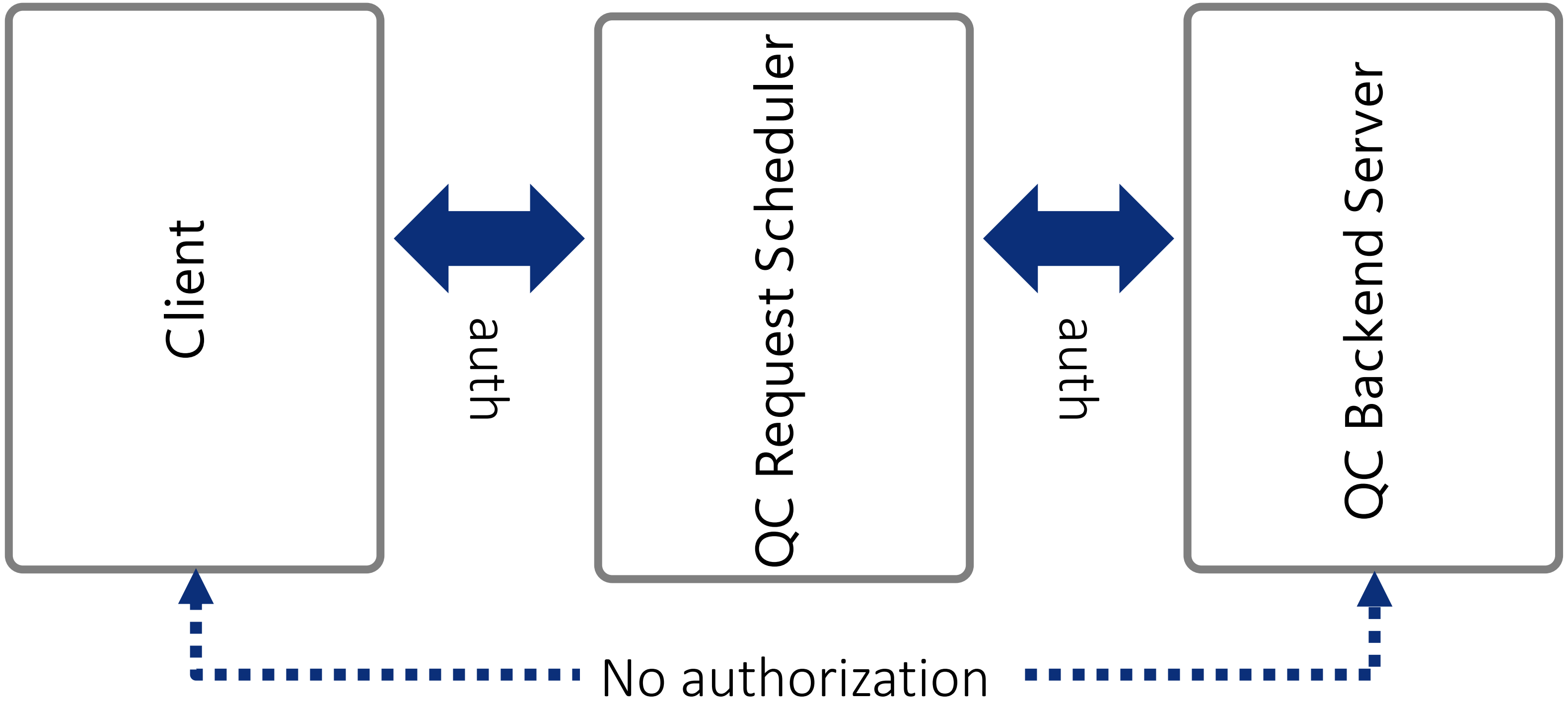
- `sqcIbmdTranspileInfo(circ, backend);`
  - User sends a request to SQC Request Scheduler to get backend property and configuration
  - results will be stored at `circ->backend_config_json` and `circ->backend_props_json`



- `sqcTranspile(circ, backend, topt);`
  - `SQC Library` transpiles a circuit by calling Qiskit python library internally
- `Users can provide the above QPU data to their own transpilers and tanspile circuits`

Authorization

- Client⇔SQC Scheduler
  - JSON Web Token (JWT) ＋TLS client certification
    - At the beginning of the session, JWT authentication is carried out over a TLS server-authenticated channel to guarantee communication confidentiality and integrity.
- SQC Scheduler ⇔ Backend Server for quantum computers
  - Authorization method specified by each quantum vendor
  - No authorization for end users



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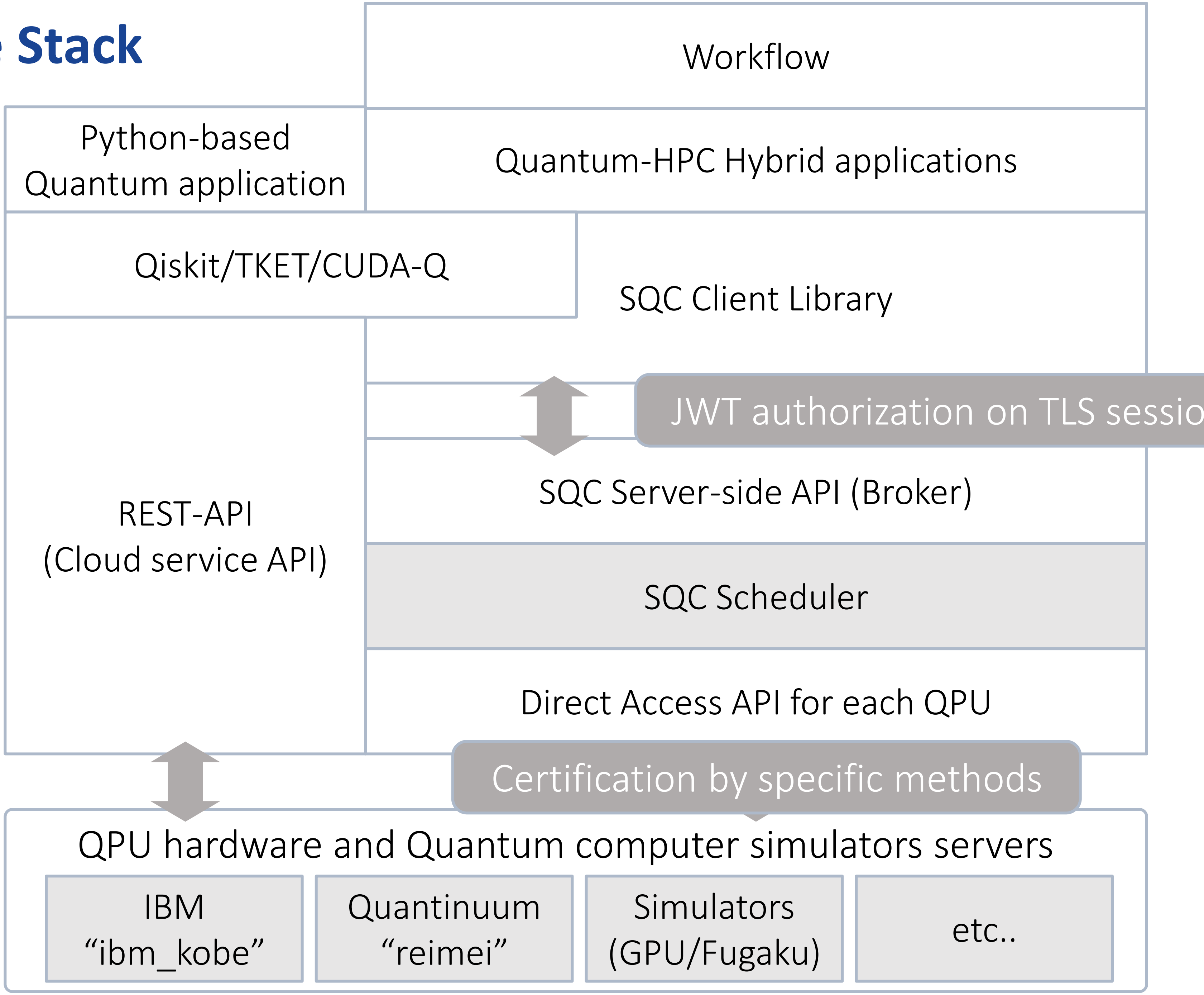
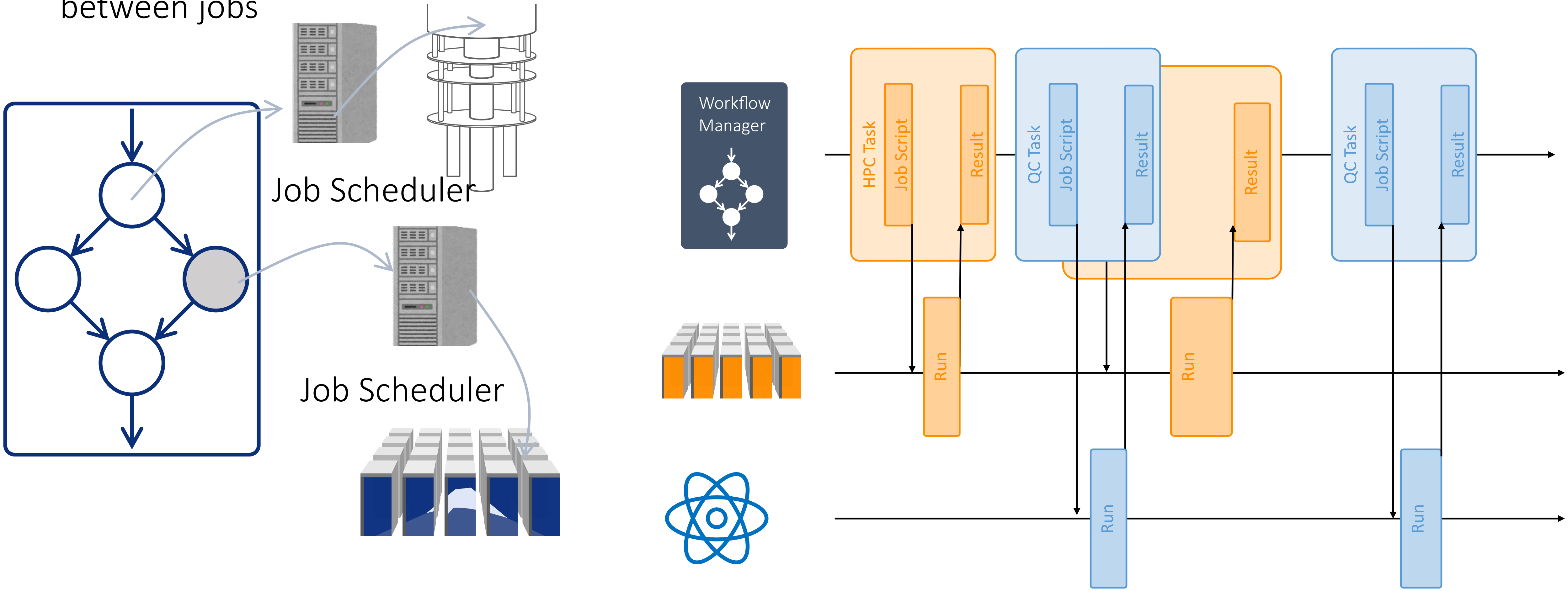


Quantum-HPC Workflow: Coarse Grained Task-based Programming

- Tasks are large, typically, whole programs, jobs in supercomputers or quantum computers
- A workflow manager will take care about the order of job submissions based on the dependency between jobs



Fine Grained Task based Programming: Example of C/C++ Program on SQC library

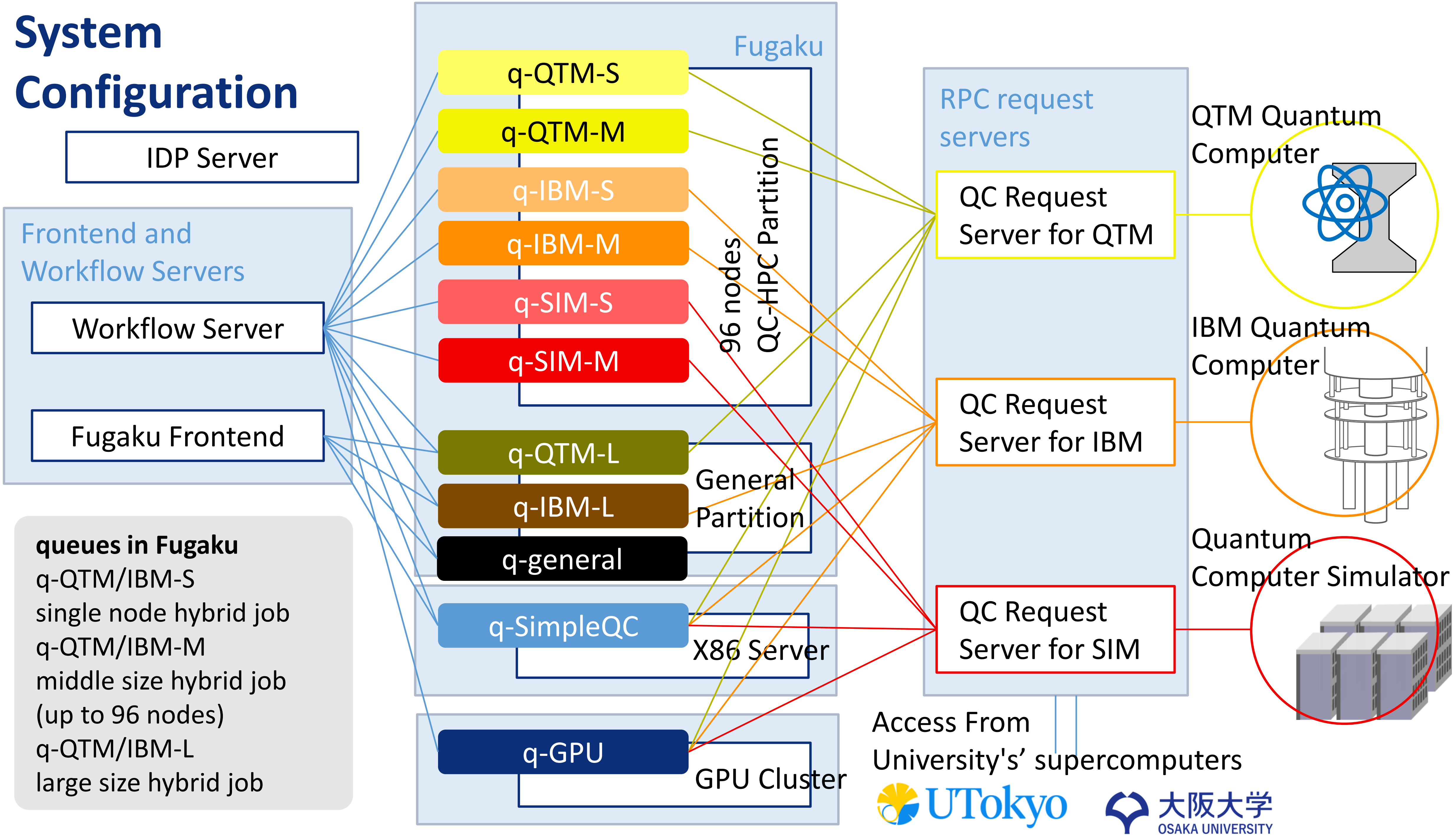
```
#include<stdio.h>
...
#include"sqc_api.h"

int main(int argc, char **argv)
{
   const int nqubits = 10;
  sqcInitOptions         *iopt;
  iopt = sqcMallocInitOptions();
  sqcInitialize(iopt);

  sqcQC* circ;
  circ = sqcQuantumCircuit(nqubits);
  sqcHGate (circ, 1);
  sqcCXGate(circ, 0, 1);
  for(int i=0; i<nqubits; i++)
    sqcMeasure(circ, i, i, NULL);

  sqcBackend backend = SQC_RPC_SCHED_QC_TYPE_QTM_SIM_GRPC;
  sqcOut result;

  circ->qasm = (char *)malloc((circ->ngates)*128);
  int len = sqcConvQASMtoMemory(circ, backend,
                                     circ->qasm, len);
  if(len <= 0) { something;}

  sqcTranspile(circ, backend, &ropt);

  sqcRunOptions         ropt;
  sqcInitializeRunOpt(&ropt);
  ropt.nshots = 30;
  ropt.qubits = nqubits;

  int ret = sqcQCRun(circ, backend, ropt, &result);

  sqcPrintQCResult(stdout, &result, ropt.outFormat);
  sqcFreeOut(&result, ropt.outFormat);

  sqcDestroyQuantumCircuit(circ);
  sqcFinalize(iopt);
  sqcFreeInitOptions(iopt);
   ....*
```

Example of Qiskit

```
from qiskit.transpiler.preset_passmanagers import
generate_preset_pass_manager
from qiskit_ibm_runtime.utils.backend_converter
import convert_to_target
from qiskit_ibm_runtime.models import
BackendProperties, BackendConfiguration
from qiskit.circuit import QuantumCircuit
from qiskit_sqc_runtime import SQCBackend,
SQCSamplerV2
from qiskit import qasm3

import json
import codecs

# Construct circuit
qc = QuantumCircuit(3, 3)
qc.h(0)
qc.cx(0, 1)
qc.cx(0, 2)
qc.measure(0,0)
qc.measure(1,1)
qc.measure(2,2)


# Get backend
backend = SQCBackend("ibm-dacc")
```

```
# Run quantum circuit
sampler = SQCSamplerV2(backend)
transpile_info = backend.transpile_info()

prop_json = json.loads(transpile_info[0])
backend_prop =
BackendProperties.from_dict(prop_json)
conf_json = json.loads(transpile_info[1])
backend_conf =
BackendConfiguration.from_dict(conf_json)

target = convert_to_target(backend_conf,
backend_prop)
pm = generate_preset_pass_manager(
  optimization_level=1,
  target=target,
)
isa_circuit = pm.run(qc)

job = sampler.run(isa_circuit, shots=10)

# Show result
result = job.result()
print(result)
print(job.status())
```

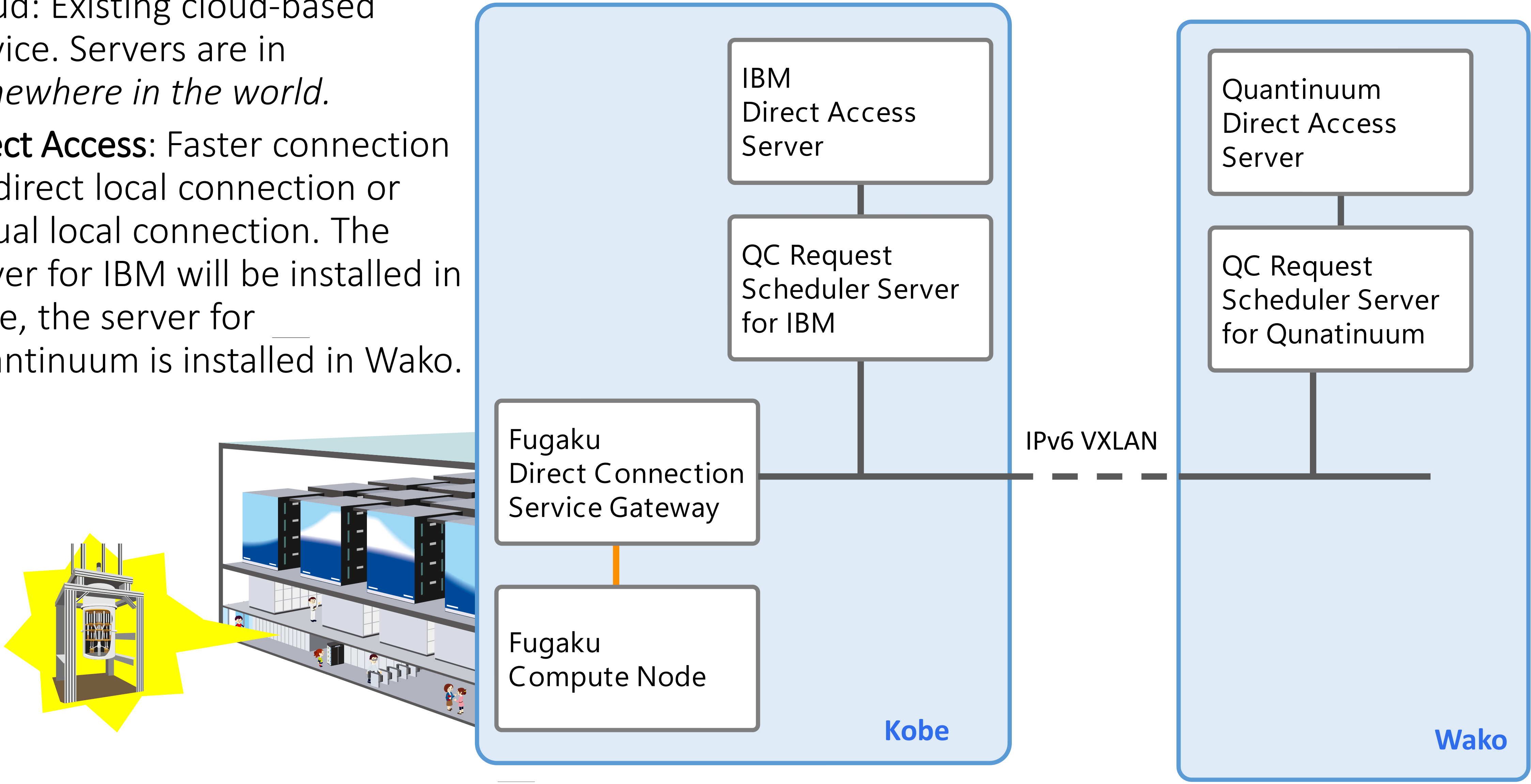
System Configuration



queues in Fugaku
q-QTM/IBM-S
single node hybrid job
q-QTM/IBM-M
middle size hybrid job
(up to 96 nodes)
q-QTM/IBM-L
large size hybrid job

QC Request Scheduler ⇔ Quantum Computer Backend Server

- Quantum Computer Backend Server
  - Cloud: Existing cloud-based service. Servers are in *somewhere in the world.*
  - **Direct Access**: Faster connection via direct local connection or virtual local connection. The server for IBM will be installed in Kobe, the server for Quantinuum is installed in Wako.



Quantum-HPC Hybrid Applications

- We are focusing on the following application areas for “Utility-Scale” Quantum-HPC hybrid computing
  - Quantum Chemistry
  - Quantum Machine Learning

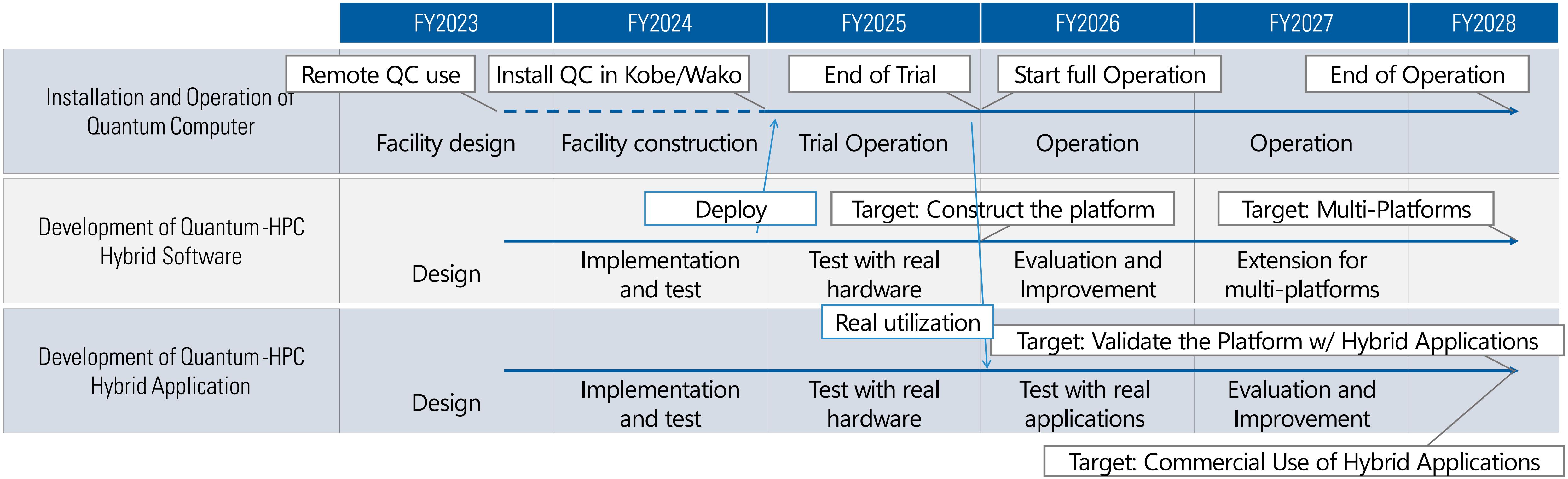
- Two experiences for Quantum-HPC Quantum Chemistry applications
  - **Ground state energy calculation of [4Fe-4S] clusters by Sample-based quantum diagonalization (SQD)**
    - Effort for Quantum-centric Supercomputing (QCSC) with IBM and RIKEN
    - SQD by Herion QPU and a large number of nodes of Fugaku (150K nodes)
    - Tightly-Coupled programming models and diagonalization by large-number of nodes.
  - **Biomolecular chemical reactions by QC-HPC hybrid computing workflow**
    - Effort on Biochemistry with Quantinuum and RIKEN
    - Loosely coupled programming models by Quantinuum TierKreis workflow tools
    - Photo-isomerization of retinal. Use NTChem developed by R-CCS executed in Fugaku for environment and QC for reaction centers

More Applications: Test User Program

- Call for application proposals to utilize the hybrid platform to research and develop quantum-HPC hybrid applications
- To obtain feedbacks on our hybrid platform
- 21 proposals are accepted
- 9 MOUs are concluded

User	Application area
JSR Corporation	Materials
Toyota Motor Corporation	Materials, Design & Manufacturing
SoftBank Corp.	Materials
Ochanomizu University	Natural Sciences
Toyota Central R&D Labs., Inc.	Design & Manufacturing
Kyoto University	Medical, Drug Discovery
Oita University	Medical, Drug Discovery
The University of Electro-Communications	Natural Sciences
Mitsubishi Chemical Corporation	Materials
and more ..	

Schedule


FY2023
FY2024
FY2025
FY2026
FY2027
FY2028
Installation and Operation of Quantum Computer
Remote QC use
Install QC in Kobe/Wako
End of Trial
Start full Operation
End of Operation
Facility design
Facility construction
Trial Operation
Operation
Operation
Development of Quantum-HPC Hybrid Software
Deploy
Target: Construct the platform
Target: Multi-Platforms
Design
Implementation and test
Test with real hardware
Evaluation and Improvement
Extension for multi-platforms
Real utilization
Development of Quantum-HPC Hybrid Application
Target: Validate the Platform w/ Hybrid Applications
Design
Implementation and test
Test with real hardware
Test with real applications
Evaluation and Improvement
Target: Commercial Use of Hybrid Applications

Summary

- Four Japanese Quantum HPC projects are ongoing
- Some details about JHPC-Quantum project
  - Programming model: Workflow for Loosely coupled programs and Tightly coupled tasks
  - Software stack
  - Scheduler between quantum and HPC
  - Use cases
    - Ground state energy calculation of [4Fe-4S] clusters by Sample-based quantum diagonalization
    - Biomolecular chemical reactions by quantum-HPC hybrid computing workflow
    - Test user program

Acknowledgment

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- This work is based on results obtained from a project, JPNP20017, commissioned by the New Energy and Industrial Technology Development Organization (NEDO).